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Regularity theory in Orlicz spaces for elliptic equations in Reifenberg domains

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Abstract

In this paper, we establish the global estimates for divergence form elliptic equations in the Orlicz space $L^{\phi}(\Omega)$, where Ω is a bounded Reifenberg domain in $\mathbb{R}^n$. Our result generalizes the $W^{1,p}$ estimates.
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1. Introduction

The aim of this paper is to study the Dirichlet problem for the divergence form elliptic equation

$$\begin{cases} -\operatorname{div}(A\nabla u) = \operatorname{div}\mathbf{f} & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{1.1}$$

Besides the uniformly elliptic assumption on the coefficient matrix $A = (a_{ij})_{n\times n}$, we also need that A satisfies the following small BMO seminorm assumption.

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Definition 1.1. The matrix A is called (δ, R)-*vanishing* if

$$\sup_{0<r\leqslant R}\sup_{x\in\mathbb{R}^n}\sqrt{\frac{1}{|B_r|}\int_{B_r(x)}\left|A(y)-\overline{A}_{B_r(x)}\right|^2 dy}\leqslant\delta, \tag{1.2}$$

where $B_r(x)$ is the open ball centered at x with radius r and $B_r = B_r(0)$, $\overline{A}_{B_r(x)} = \frac{1}{|B_r|}\int_{B_r(x)} A(y)\,dy$.

As to the domain Ω, we assume that $\Omega\subseteq\mathbb{R}^n$ is bounded, open and (δ, R)-Reifenberg flat. More precisely,

Definition 1.2. We say a domain Ω is (δ, R)-*Reifenberg flat* if for every $x\in\partial\Omega$ and every $r\in(0, R]$, there exists an $(n-1)$-dimensional plane $L(x, r)$ such that

$$\frac{1}{r}D\big[\partial\Omega\cap B_r(x), L(x,r)\cap B_r(x)\big]\leqslant\delta, \tag{1.3}$$

where $D[A, B] = \sup\{\operatorname{dist}(x, B)\colon x\in A\}+\sup\{\operatorname{dist}(y, A)\colon y\in B\}$ is the Hausdorff distance of A and B.

Throughout this paper we mean $\delta>0$ to be a small constant since Definition 1.2 is only significant for $\delta>0$ small. The Reifenberg flat domain is locally like a topological disk of proper dimension. They may be fractal, but they are $W^{1,p}$ $(1\leqslant p\leqslant\infty)$ extension domains. They are geometrically invariant with respect scaling. These domains appear in applied mathematics such as zigzag domains, domains of costal lines and domains of atomic clusters. For more about Reifenberg domains, we refer the readers to [6,8].

In the Orlicz space, a lot of work about regularity theory of linear and nonlinear elliptic equations has been done [1,2,4,5,9]. Azroul et al. [2] proved radial solution u to the Poisson equation satisfies $\frac{\partial^2 u}{\partial x_i\partial x_j}\in L^{\phi}_{\mathrm{loc}}(B_R)$ if $f\in L^{\phi}(B_R)$ and $\phi(|f(x)|)\log(|x|)$ are integrable. The strongly nonlinear elliptic problem in the Orlicz space L^{ϕ} is solvable with the assumption that the Young function ϕ satisfies Δ_2 condition (see Definition 2.4) near infinity [4]. Moreover, it has been proved in [10] that the Riesz transform R_j is bounded in L^{ϕ} if and only if $\phi\in\Delta_2\cap\nabla_2$ (see Definitions 2.4 and 2.5). Recently, Byun and Wang [3] investigated a suitable and minimal condition on the domain for the $W^{1,p}$ regularity theory with $1<p<\infty$. They showed that

Theorem 1.1. *(See [3].) Let $1<p<\infty$. Then there exists a small $\delta=\delta(\Lambda, p, n, R)>0$ so that for all A with A uniformly elliptic and (δ, R)-vanishing, for all Ω with Ω (δ, R)-Reifenberg flat, and for all* $\mathbf{f}$ *with* $\mathbf{f}\in L^p(\Omega;\mathbb{R}^n)$*, the Dirichlet problem* (1.1) *has s unique weak solution with the estimate*

$$\int_{\Omega}|\nabla u|^p\leqslant C\int_{\Omega}|\mathbf{f}|^p\,dx. \tag{1.4}$$

All of the mentioned work motivates us to study the integrability of the gradient of the solution in the setting of Orlicz spaces. The main goal of this paper is to extend the result of [3] from L^p to the general Orlicz space L^{ϕ}. We shall prove

$$\int_{\Omega}\phi(Du)\,dx\leqslant C\int_{\Omega}\phi(\mathbf{f})\,dx \tag{1.5}$$

holds with some assumptions on ϕ, see Theorem 3.1. In the context, C is the constant independent of u and $\mathbf{f}$, which may have different values in different occurrences.

Our approach to derive (1.5) is similar to that in [3] and [13], it is based on an approximation argument, the modified Vitali covering lemma and a compactness method. In Section 2 we introduce some definitions and analysis results, mainly concerning Orlicz spaces and maximal functions. In Section 3 we discuss the global regularity for the Dirichlet problem (1.1).

2. Preliminaries

In this section, we recall some definitions and basic facts about Orlicz spaces. For an exhaustive treatment of the theory of Orlicz and Orlicz–Sobolev spaces, we refer the readers to [11]. Some analysis results are also listed (see Refs. [7,10,12]).

We need to introduce some notations here.

(1) $B_r(x) = \{y \in \mathbb{R}^n\colon\ |y - x| < r\}$ is an open ball centered at x with radius r. $B_r = B_r(0)$, $B_r^+ = B_r \cap \{x_n > 0\}$.
(2) $T_r = B_r \cap \{x_n = 0\}$, $T_r(x) = T_r + x$ is the transition of T_r.
(3) $\Omega_r = \Omega \cap B_r$, $\Omega_r(x) = \Omega \cap B_r(x)$.
(4) $\partial\Omega_r$ is the boundary of Ω_r. $\partial_w\Omega_r = \partial\Omega \cap B_r(x)$ is the wiggled part of $\partial\Omega_r$, $\partial_c\Omega_r = \partial\Omega_r \setminus \partial_w\Omega_r$ is the curved part of $\partial\Omega_r$.
(5) A matrix $A = (a_{ij})_{n\times n}$ is uniformly elliptic if there exist positive constants λ and Λ such that

$$\lambda|\xi|^2 \leqslant A(x)\xi\cdot\xi \leqslant \Lambda|\xi|^2 \quad \text{a.e. } x \in \mathbb{R}^n,\ \forall \xi \in \mathbb{R}^n.$$

(6) $|A| = \sqrt{\sum_{i,j=1}^n a_{ij}^2}$, $\|A\|_\infty = \sup_y |A(y)|$.
(7) $\chi_\Omega(x)$ is the characteristic function of Ω. $\chi_\Omega(x) = 1$ if $x \in \Omega$ and $\chi_\Omega(x) = 0$ outside Ω.
(8) For a locally integrable function f,

$$\bar{f}_{B_r(x)} = \frac{1}{|B_r|}\int\limits_{B_r(x)} f(y)\,dy$$

is the average of f over $B_r(x)$.
(9) For a measurable function f, define its distribution function

$$\mu_f(\lambda) = \big|\big\{x \in \Omega\colon\ |f| > \lambda\big\}\big|.$$

We denote by Φ the function class that consists of all functions $\phi\colon R^1 \to [0,\infty)$ which are nonnegative, even, nondecreasing on $[0,\infty)$ and satisfy $\phi(0+) = 0$, $\lim_{t\to\infty}\phi(t) = \infty$.

Definition 2.1. A function $\phi \in \Phi$ is said to be a *Young function* if ϕ is convex and $\lim_{t\to 0^+}\phi(t)/t = \lim_{t\to\infty} t/\phi(t) = 0$.

Let $\Omega \subseteq \mathbb{R}^n$ be a bounded open set, ϕ be a Young function. The *Orlicz class* $K^\phi(\Omega)$ is the set of all measurable functions $f\colon \Omega \to \mathbb{R}^1$ satisfying

$$\int\limits_\Omega \phi\big(f(x)\big)\,dx < \infty. \tag{2.1}$$

Definition 2.2. The *Orlicz space* $L^{\phi}(\Omega)$ is defined to be the linear hull of $K^{\phi}(\Omega)$. It consists of all measurable functions f such that $\lambda f \in K^{\phi}(\Omega)$ for some $\lambda > 0$.

Definition 2.3. Let ϕ be a Young function, the *Sobolev–Orlicz space* $W^1L^{\phi}(\Omega)$ is the space of all functions f such that the distributional derivatives $D^{\alpha} f \in L^{\phi}$ with $|\alpha| \leqslant 1$. The space $W_0^1 L^{\phi}(\Omega)$ is defined as the closure of the Schwartz space $\mathcal{D}(\Omega)$ in $W^1L^{\phi}(\Omega)$.

For $\phi \in \Phi$, we define the complementary function (with respect to ϕ) by

$$\tilde{\phi}(s) = \sup\bigl\{st - \phi(t);\ t \geqslant 0\bigr\}. \tag{2.2}$$

It is obvious from the definition that $\tilde{\phi}$ is a Young function if ϕ is a one. They satisfy the Young inequality

$$ts \leqslant \phi(t) + \tilde{\phi}(s), \quad t, s \geqslant 0. \tag{2.3}$$

Definition 2.4. A Young function ϕ is said to satisfy the *global* Δ_2 *condition* (Δ_2 condition near infinity), denoted by $\phi \in \Delta_2$ ($\phi \in \Delta_2$ near infinity), if there exists $K > 0$ such that $\phi(2t) \leqslant K\phi(t)$ for all $t > 0$ ($t \geqslant t_0 > 0$).

Actually the Δ_2 condition for ϕ can be formulated in the following way: for every $A > 1$, there exists $\beta = \log_2 K$ such that $\phi(At) \leqslant KA^{\beta}\phi(t)$ for all $t > 0$.

Definition 2.5. A Young function ϕ is said to satisfy the *global* ∇_2 *condition*, denoted by $\phi \in \nabla_2$, if there exists $a > 1$ such that for all $t > 0$,

$$\phi(t) \leqslant \frac{\phi(at)}{2a}. \tag{2.4}$$

In fact, $\phi \in \nabla_2$ implies that for any $0 < \lambda < 1$, there exists an integer $m \geqslant 0$ such that $\frac{1}{a^{m+1}} \leqslant \lambda < \frac{1}{a^m}$. Then

$$\phi(\lambda t) \leqslant \phi\left(\frac{t}{a^m}\right) \leqslant \frac{2a\phi(t)}{(2a)^{m+1}} = \frac{2a\phi(t)}{(a^{m+1})^{1+\log_a 2}}.$$

That is, for any $0 < \lambda < 1$, $\phi(\lambda t) \leqslant 2a\lambda^{\gamma}\phi(t)$ with $\gamma = \log_a 2 + 1$.

It should be noted that both Δ_2 and ∇_2 conditions make the function ϕ grow moderately. For instance, $\phi(t) = |t|^p(1 + |\log|t||) \in \Delta_2 \cap \nabla_2$ for $p > 1$.

Remark 2.1. In general, $K^{\phi} \subset L^{\phi}$. If $\phi \in \Delta_2$, then $K^{\phi} = L^{\phi}$, C_0^{∞} is dense in L^{ϕ} and many properties of L^{ϕ} are close to properties of the standard L^p space.

Remark 2.2. If $f \in L^{\phi}$, then $\int_{\Omega} \phi(f)\,dx$ can be rewritten as an integral of the distribution function $\mu_f(\lambda)$. That is, for $N > 1$,

$$\int_{\Omega} \phi(f)\,dx = \sum_{k=-\infty}^{\infty} \int_{N^k}^{N^{k+1}} \mu_f(\lambda)\,d\phi(\lambda).$$

Definition 2.6. For a locally integrable function u defined on $\mathbb{R}^n$, we define its *maximal function* $(\mathcal{M}u)(x)$ as

$$(\mathcal{M}u)(x) = \sup_{r>0} \frac{1}{|B_r|} \int_{B_r(x)} \big|u(y)\big|\, dy. \tag{2.5}$$

We also use

$$(\mathcal{M}_\Omega u)(x) = (\mathcal{M}u\chi_\Omega)(x), \tag{2.6}$$

where χ_Ω is the characteristic function of Ω. We will drop the subscript Ω if Ω is clear in the context.

The maximal operator $\mathcal{M}$ is bounded in L^p for $1 < p \leqslant \infty$. More precisely,

Theorem 2.1. *(See [12].)*

$$\big|\big\{x \in \Omega\colon\ (\mathcal{M}f)(x) > \alpha\big\}\big| \leqslant \frac{C}{\alpha} \int_\Omega \big|f(x)\big|\, dx, \quad \forall \alpha > 0, \tag{2.7}$$

$$\big\|(\mathcal{M}f)\big\|_{L^p} \leqslant C\|f\|_{L^p}, \quad 1 < p \leqslant \infty. \tag{2.8}$$

The following theorem gives a characterization of those functions $\phi \in \Phi$ for which

$$\int_{\mathbb{R}^n} \phi\big(\mathcal{M}f(x)\big)\, dx \leqslant C \int_{\mathbb{R}^n} \phi\big(cf(x)\big)\, dx \tag{2.9}$$

holds for all $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$.

Theorem 2.2. *(See [10].) Let ϕ be a Young function. The following statements are equivalent*:

(i) (2.9) *is true for all* $f \in L^1_{\mathrm{loc}}$;
(ii) *there exists a positive constant* $a > 1$ *such that for any* $t > 0$,

$$\phi(t) < \frac{1}{2a}\phi(at). \tag{2.10}$$

The condition (ii) in Theorem 2.2 is equivalent to $\phi \in \nabla_2$. So $\phi \in \Delta_2 \cap \nabla_2$ implies the boundedness of the maximal operator in L^ϕ.

At the end of this section, we introduce a modified version of the Vitali covering lemma on $(\delta, 1)$-Reifenberg flat domain. See [3] for the proof of this lemma.

Lemma 2.1. *(See [3].) Assume that G and H are two bounded measurable sets, $G \subset H \subset \Omega$ with Ω $(\delta, 1)$-Reifenberg flat. There exists a $\varepsilon > 0$ such that for all $x \in \Omega$, $|G \cap B_r(x)| \geqslant \varepsilon |B_r(x)|$ implies that $B_r(x) \cap \Omega \subset H$. Then*

$$|G| \leqslant \left(\frac{10}{1-\delta}\right)^n \varepsilon |H|. \tag{2.11}$$

3. Global estimates for elliptic equations in Reifenberg domain in Orlicz space

In this section, we consider the global regularity of weak solutions to (1.1) in the Orlicz space L^ϕ. Analogue to [3], a compactness method and a modified Vitali covering lemma will be applied. The main theorem of this paper is as follows.

Theorem 3.1. *Let $\phi \in \Delta_2 \cap \nabla_2$. Then there exists a small $\delta = \delta(n, \phi, \Lambda, R)$, such that for all A with A uniformly elliptic and (δ, R) vanishing, for all Ω with Ω (δ, R)-Reifenberg flat, and for all $\mathbf{f}$ with $\mathbf{f} \in L^\phi(\Omega; \mathbb{R}^n)$, the Dirichlet problem* (1.1) *has a unique weak solution $u \in W_0^1 L^\phi(\Omega)$. Moreover,*

$$\int_\Omega \phi(Du)\,dx \leqslant C_1 \int_\Omega \phi(\mathbf{f})\,dx, \tag{3.1}$$

where C_1 depends on Ω, n and ϕ.

Here we need the definition of weak solutions in $W_0^1 L^\phi$.

Definition 3.1. Let $(\phi, \tilde{\phi})$ be a complementary pair of Young functions. $u \in W_0^1 L^\phi(\Omega)$ is called a weak solution of problem (1.1) if for any $\psi \in W_0^1 L^{\tilde{\phi}}(\Omega)$,

$$\int_\Omega A\nabla u \cdot \nabla\psi\,dx = -\int_\Omega \mathbf{f} \cdot \nabla\psi\,dx.$$

To overcome the difficulties arising from the lack of regularity of $\partial\Omega$, we shall locally approximate the solution of (1.1) by a solution to a suitable homogeneous problem. Namely, we approximate any weak solution u of

$$\begin{cases} -\operatorname{div}(A\nabla u) = \operatorname{div}\mathbf{f} & \text{in } \Omega_r, \\ u = 0 & \text{on } \partial_w\Omega_r \end{cases} \tag{3.2}$$

by a weak solution v to

$$\begin{cases} -\operatorname{div}(\widetilde{A}\nabla v) = 0 & \text{in } B_r^+, \\ v = 0 & \text{on } T_r, \end{cases} \tag{3.3}$$

where $\widetilde{A}$ is a suitably chosen constant matrix with $\|\overline{A}_{B_r^+} - \widetilde{A}\|_\infty$ small.

In the sequel, the condition $B_r \cap \{x_n > -\delta\} \supset \Omega_r \supset B_r^+$ will be used usually. It means that $\Omega_r \supset T_r$ with $D(\partial\Omega_r, T_r)$ small, where D denotes the Hausdorff distance.

Lemma 3.1. *For any $1 < q < \infty$, $\varepsilon > 0$, there exists a small $\delta = \delta(\varepsilon, q) > 0$ such that for any weak solution u of* (3.2) *in Ω_4 with*

$$B_4 \cap \{x_n > -\delta\} \supset \Omega_4 \supset B_4^+$$

and

$$\frac{1}{|B_4|}\int_{\Omega_4} |\nabla u|^q\,dx \leqslant 1, \qquad \frac{1}{|B_4|}\int_{\Omega_4} \left(|f|^q + |A - \overline{A}_{\Omega_4}|^q\right)dx \leqslant \delta^q, \tag{3.4}$$

there exist a constant matrix $\widetilde{A}$ *with* $\|\overline{A}_{B_4^+} - \widetilde{A}\|_{L^\infty} \leqslant \varepsilon$ *and a corresponding weak solution of* (3.3) *in* B_4^+ *such that*

$$\int\limits_{B_4^+} |u - v|^q \, dx \leqslant \varepsilon^q. \tag{3.5}$$

Proof. We prove this lemma by contradiction. If not, there exist $\varepsilon_0 > 0$, $\{A_k\}_{k=1}^\infty$, $\{u_k\}_{k=1}^\infty$, $\{\mathbf{f}_k\}_{k=1}^\infty$ and $\{\Omega_4^k\}_{k=1}^\infty$ such that u_k is a weak solution of

$$\begin{cases} -\operatorname{div}(A_k \nabla u_k) = \operatorname{div} \mathbf{f}_k & \text{in } \Omega_4^k, \\ u_k = 0 & \text{on } \partial_w \Omega_4^k \end{cases} \tag{3.6}$$

with

$$B_4 \cap \left\{ x_n > -\frac{1}{k} \right\} \supset \Omega_4^k \supset B_4^+$$

and

$$\frac{1}{|B_4|} \int\limits_{\Omega_4^k} |\nabla u_k|^q \, dx \leqslant 1, \qquad \frac{1}{|B_4|} \int\limits_{\Omega_4^k} \left(|\mathbf{f}_k|^q + |A_k - \overline{A}_{k\Omega_4^k}|^q \right) dx \leqslant \frac{1}{k^q}. \tag{3.7}$$

But

$$\int\limits_{B_4^+} |u_k - v|^q \, dx > \varepsilon_0^q \tag{3.8}$$

for any v satisfies (3.3) in B_4^+ with $\|\overline{A}_{kB_4^+} - \widetilde{A}\|_{L^\infty} \leqslant \varepsilon_0$.

Since $u_k = 0$ on $\partial_w \Omega_4$, $\{u_k\}_{k=1}^\infty$ is bounded in $W^{1,q}$ from (3.7). There exists a subsequence, which still be denoted as $\{u_k\}_{k=1}^\infty$, and $u_0 \in W^{1,q}(B_4^+)$ such that

$$u_k \rightharpoonup u_0 \quad \text{in } W^{1,q}\left(B_4^+\right), \qquad u_k \to u_0 \quad \text{in } L^q\left(B_4^+\right). \tag{3.9}$$

Since $\{\overline{A}_{kB_4^+}\}$ is bounded in L^∞, it has a subsequence, denoted as $\overline{A}_k$, satisfies that

$$\|\overline{A}_k - A_0\|_{L^\infty} \to 0 \quad \text{as } k \to \infty \tag{3.10}$$

for some constant matrix A_0. From (3.10) and (3.7), there is a subsequence of A_k, denoted also as A_k,

$$A_k \to A_0 \quad \text{in } L^q\left(B_4^+\right). \tag{3.11}$$

If we show that u_0 is a weak solution of

$$\begin{cases} -\operatorname{div}(A_0 \nabla u_0) = 0 & \text{in } B_4^+, \\ u_0 = 0 & \text{on } T_4, \end{cases} \tag{3.12}$$

then we get a contradiction to (3.8) by taking $\widetilde{A} = A_0$, $v = u_0$ and k large enough.

To prove that u_0 is a weak solution of (3.12), we need to show:

(1) $\int_{B_4^+} A_0 \nabla u_0 \nabla \psi \, dx = 0$ for any $\psi \in W_0^{1,q^*}(B_4^+)$ with $\frac{1}{q} + \frac{1}{q^*} = 1$.
(2) $u_0 = 0$ on T_4.

Firstly, we extend ψ by 0 outside B_4^+. From (3.6),

$$\int\limits_{B_4^+} A_k \nabla u_k \nabla \psi \, dx = -\int\limits_{B_4^+} \mathbf{f}_k \nabla \psi \, dx. \tag{3.13}$$

Combining (3.9) with (3.11), we have

$$A_k \nabla u_k \rightharpoonup A_0 \nabla u_0 \quad \text{in } L^q\big(B_4^+\big) \text{ as } k \to \infty. \tag{3.14}$$

Let $k \to \infty$ in (3.13), we find that $\int_{B_4^+} A_0 \nabla u_0 \nabla \psi \, dx = 0$.

Now we take $x' \in T_4$, $\theta > 0$ fixed. Set $s_0 = \min\{s\colon\ \partial\Omega_4^k \cap (x', \theta - s) \neq \emptyset\}$, then $0 < s_0 \leqslant \theta + \frac{1}{k}$.

Take S as the closure of $\{w \in C^\infty(\overline{B}_4)\colon\ w = 0 \text{ on } \partial_w \Omega_4^k\}$ in $W^{1,q}(B_4)$, then $u_k \in S$. Without loss of generality, we may assume that $u_k \in C^\infty(\overline{B}_4)$ and $u_k(x', \theta - s_0) = 0$. Then

$$\begin{aligned}
\big|u_k(x', \theta)\big|^q &= \big|u_k(x', \theta) - u_k(x', \theta - s_0)\big|^q \\
&= s_0^q \left| \int\limits_0^1 \frac{\partial}{\partial x_n} u_k\big(x', \theta - (1-s)s_0\big)\, ds \right|^q \\
&\leqslant 2^q \left(\theta^q + \frac{1}{k^q}\right) \int\limits_0^1 \big|\nabla u_k\big(x', \theta - (1-s)s_0\big)\big|^q \, ds.
\end{aligned}$$

Integrating the above inequality with respect to x' over T_4, we derive

$$\begin{aligned}
\int\limits_{T_4} \big|u_k(x', \theta)\big|^q \, dx' &\leqslant 2^q \left(\theta^q + \frac{1}{k^q}\right) \int\limits_{T_4} \int\limits_0^1 \big|\nabla u_k\big(x', \theta - (1-s)s_0\big)\big|^q \, ds \, dx' \\
&\leqslant 2^q \left(\theta^q + \frac{1}{k^q}\right) \int\limits_{\Omega_4^k} |\nabla u_k|^q \, dx \\
&\leqslant C\left(\theta^q + \frac{1}{k^q}\right).
\end{aligned}$$

Let $k \to \infty$ and $\theta \to 0$, we obtain

$$\int\limits_{T_4} \big|u_0(x', 0)\big|^q \, dx' = 0.$$

Thus we complete the proof of this lemma. □

Corollary 3.1. *For any $1 < q < n$, $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon, q) > 0$ such that the following statement holds*: *if u is a weak solution of* (3.2) *in Ω_4 with*

$$B_4 \cap \{x_n > -\delta\} \supset \Omega_4 \supset B_4^+ \tag{3.15}$$

and

$$\frac{1}{|B_4|} \int\limits_{\Omega_4} |\nabla u|^q \, dx \leqslant 1, \qquad \frac{1}{|B_4|} \int\limits_{\Omega_4} \big(|f|^q + |A - \overline{A}_{\Omega_4}|^q\big) \, dx \leqslant \delta^q, \tag{3.16}$$

then there exist a constant matrix $\widetilde{A}$ *with* $\|\overline{A}_{B_4^+} - \widetilde{A}\|_\infty \leqslant \varepsilon$ *and a corresponding weak solution of* (3.3) *in* B_4^+ *such that*

$$\int\limits_{\Omega_2} \left|\nabla(u-V)\right|^q dx \leqslant \varepsilon^q, \tag{3.17}$$

where V *is a* 0*-extension of* v *defined in* B_4^+ *to* B_4.

Proof. From Lemma 3.1, for any $\eta > 0$, there exist a small $\delta(\eta, q)$, a constant matrix $\widetilde{A}$ with $\|\overline{A}_{B_4^+} - \widetilde{A}\|_{L^\infty} \leqslant \eta$, and a corresponding solution v of (3.3) in B_4^+ such that

$$\int\limits_{B_4^+} |u-v|^q \leqslant \eta^q. \tag{3.18}$$

Since V is a 0-extension of v defined in B_4^+ to B_4, we claim that V is a weak solution of

$$\begin{cases} -\operatorname{div}\big(\widetilde{A}\nabla V(x)\big) = -\dfrac{\partial}{\partial x_n}\left(\tilde{a}_{nn}\dfrac{\partial v}{\partial x_n}(x',0)\chi_{\mathbb{R}^n_-}(x)\right) & \text{in } \Omega_4, \\ V = 0 \quad \text{on } \partial_w\Omega_4. \end{cases}$$

Then $u - V \in W^{1,q}(\Omega_4)$ is a weak solution of

$$-\operatorname{div}\big(A\nabla(u-V)\big) = \operatorname{div}\big(\mathbf{f} + (A-\widetilde{A})\nabla V\big) + \frac{\partial}{\partial x_n}\left(\tilde{a}_{nn}\frac{\partial v}{\partial x_n}(x',0)\chi_{\mathbb{R}^n_-}(x)\right)$$

in Ω_4 with the boundary condition $u - V = 0$ on $\partial_w\Omega_4$. From Theorem 1.1, we have the following estimate:

$$\begin{aligned} \big\|\nabla(u-V)\big\|^q_{L^q(\Omega_2)} &\leqslant C\left(\big\|\mathbf{f} + (A-\widetilde{A})\nabla V\big\|^q_{L^q(\Omega_4)}\right. \\ &\qquad \left. + \left\|\tilde{a}_{nn}\frac{\partial v}{\partial x_n}(x',0)\chi_{\mathbb{R}^n_-}(x)\right\|^q_{L^q(\Omega_4)} + \|u-V\|^q_{L^q(\Omega_4)}\right) \\ &\leqslant C\big(\delta^q + \|u-V\|^q_{L^q(\Omega_4)}\big). \end{aligned}$$

From the Hölder inequality and the assumption on the domain, we have

$$\int\limits_{\Omega_4\setminus B_4^+} |u-V|^q\,dx \leqslant \left(\int\limits_{\Omega_4\setminus B_4^+} |u-V|^{\frac{nq}{n-q}}\,dx\right)^{\frac{n-q}{n}} \big|\Omega_4 \setminus B_4^+\big|^{\frac{q}{n}} \leqslant C\delta^{\frac{q}{n}}.$$

Then

$$\begin{aligned} \int\limits_{\Omega_4} |u-V|^q\,dx &= \int\limits_{\Omega_4\setminus B_4^+} |u-V|^q\,dx + \int\limits_{B_4^+} |u-V|^q\,dx \\ &\leqslant C\left(\delta^{\frac{q}{n}} + \int\limits_{B_4^+} |u-v|^q\,dx\right) \\ &\leqslant C\big(\delta^{\frac{q}{n}} + \eta^q\big). \end{aligned}$$

Hence

$$\int\limits_{\Omega_2} \left|\nabla(u-V)\right|^q dx \leqslant C\left(\delta^{\frac{q}{n}} + \eta^q\right) = \varepsilon \tag{3.19}$$

by taking η and δ suitable in the inequality above. Thus the proof is completed. □

Now, let us show the key lemma of this paper. It is the so called "good-λ" inequality. It says that the existence of one "good point" implies that most points are "good."

Lemma 3.2. *There is a constant $N_1 > 1$ depending only on n, such that for any $\varepsilon > 0$, $1 < q < n$, there exists a small $\delta = \delta(\varepsilon, q) > 0$ with the following argument holds: Suppose that A is uniformly elliptic and $(\delta, 6)$-vanishing, u is a weak solution of* (3.2) *in Ω with $B_6 \cap \{x_n > -\delta\} \supset \Omega_6 \supset B_6^+$. If*

$$\left\{x \in \Omega\colon\ \mathcal{M}\left(|\nabla u|^q\right) \leqslant 1\right\} \cap \left\{x \in \Omega\colon\ \mathcal{M}\left(|\mathbf{f}|^q\right) \leqslant \delta^q\right\} \cap B_1 \neq \emptyset, \tag{3.20}$$

then

$$\left|\left\{x \in \Omega\colon\ \mathcal{M}\left(|\nabla u|^q\right) > N_1^q\right\} \cap B_1\right| \leqslant \varepsilon |B_1|. \tag{3.21}$$

Proof. By (3.20), there is a point $x_0 \in \Omega_1 = \Omega \cap B_1$, such that

$$\frac{1}{|B_r|} \int\limits_{\Omega_r(x_0)} |\nabla u|^q \, dx \leqslant 1 \quad \text{and} \quad \frac{1}{|B_r|} \int\limits_{\Omega_r(x_0)} |\mathbf{f}|^q \, dx \leqslant \delta^q$$

for all $r > 0$. Since $\Omega_4(0) \subset \Omega_5(x_0)$, we see that

$$\frac{1}{|B_4|} \int\limits_{\Omega_4(0)} |\nabla u|^q \, dx \leqslant \left(\frac{5}{4}\right)^n \quad \text{and} \quad \frac{1}{|B_4|} \int\limits_{\Omega_4(0)} |\mathbf{f}|^q \, dx \leqslant \left(\frac{5}{4}\right)^n \delta^q.$$

With u replaced by $(\frac{5}{4})^n u$ and $\mathbf{f}$ replaced by $(\frac{5}{4})^n \mathbf{f}$, we apply Corollary 3.1 to find that for any $\eta > 0$, $1 < q < n$, there exists a small $\delta(\eta, q) > 0$ and a corresponding weak solution v of (3.3) in B_4^+ such that

$$\int\limits_{\Omega_2} \left|\nabla(u-V)\right|^q \leqslant \eta^q$$

provided that

$$\frac{1}{|B_4|} \int\limits_{\Omega_4} \left(|f|^q + |A - \overline{A}_{\Omega_4}|^q\right) dx \leqslant \delta^q,$$

where V is a 0-extension of v defined in B_4^+ to B_4. By applying the local $W^{1,\infty}$ estimate for V, there is a constant N_0 such that

$$\|\nabla V\|_{L^\infty(\Omega_3)} \leqslant N_0.$$

Now we claim that

$$\left\{x \in \Omega_1\colon\ \mathcal{M}\left(|\nabla u|^q\right) > N_1^q\right\} \subset \left\{x \in \Omega_1\colon\ \mathcal{M}_{\Omega_4}\left(\left|\nabla(u-V)\right|^q\right) > N_0^q\right\}, \tag{3.22}$$

where $N_1^q = \max(2^{q+1} N_0^q, 2^n)$.

Take $x \in \{x \in \Omega_1 \colon \mathcal{M}_{\Omega_4}(|\nabla(u-V)|^q) \leqslant N_0^q\}$. If $r \leqslant 2$, we have $\Omega_r(x) \subset \Omega_3$ and

$$\sup_{r\leqslant 2} \frac{1}{|B_r|} \int\limits_{\Omega_r(x)} \left|\nabla u(y)\right|^q dy \leqslant 2^q \left(\mathcal{M}_{\Omega_3}\left(\left|\nabla(u-V)\right|^q\right) + N_0^q\right) \leqslant 2^{(q+1)} N_0^q.$$

For $r > 2$, $x_0 \in \Omega_r(x) \subset \Omega_{2r}(x_0)$, we have

$$\frac{1}{|B_r|} \int\limits_{\Omega_r(x)} |\nabla u|^q \, dy \leqslant \frac{1}{|B_r|} \int\limits_{\Omega_{2r}(x_0)} |\nabla u|^q \, dy \leqslant 2^n.$$

This implies that $\mathcal{M}(|\nabla u|^q)(x) \leqslant N_1^q$.

Finally, we have

$$\begin{aligned} \left|\left\{x \in \Omega_1 \colon \mathcal{M}\left(|\nabla u|^q\right)(x) > N_1^q\right\}\right| &\leqslant \left|\left\{x \in \Omega_1 \colon \mathcal{M}_{\Omega_4}\left(\left|\nabla(u-V)\right|^q\right) > N_0^q\right\}\right| \\ &\leqslant \frac{C}{N_0^q} \int\limits_{\Omega_4} \left|\nabla(u-V)\right|^q dx \leqslant C\eta^q = \varepsilon |B_1|. \end{aligned}$$

By taking δ and η suitable in the inequalities above, the proof is completed. □

The scaled version of Lemma 3.2 is as follows.

Corollary 3.2. *There is a constant $N_1 > 0$ such that for any $\varepsilon, r > 0$, $1 < q < n$, there exists a small $\delta = \delta(\varepsilon, q) > 0$ with A uniformly elliptic and $(\delta, 6r)$-vanishing. If u is a weak solution of* (3.2) *in Ω with*

$$B_{6r} \cap \{x_n > -\delta r\} \supset \Omega_{6r} \supset B_{6r}^+$$

and

$$\left\{x \in \Omega_r \colon \mathcal{M}\left(|\nabla u|^q\right) \leqslant 1\right\} \cap \left\{x \in \Omega_r \colon \mathcal{M}\left(|\mathbf{f}|^q\right) \leqslant \delta^q\right\} \neq \emptyset,$$

then

$$\left|\left\{x \in \Omega_r \colon \mathcal{M}\left(|\nabla u|^q\right) > N_1^q\right\}\right| < \varepsilon |B_r|.$$

To apply the modified Vitali covering lemma, we need the following lemma to verify the condition needed in Lemma 2.1.

Lemma 3.3. *There is a constant $N_1 > 1$ depending only on n such that for any $\varepsilon > 0$, $r \in (0, 1]$, $1 < q < n$, there exists a small $\delta = \delta(\varepsilon, q) > 0$ with A uniformly elliptic and $(\delta, 48)$-vanishing, and if u is a weak solution of* (3.2) *in Ω with Ω $(\delta, 48)$-Reifenberg flat, and*

$$\left|\left\{x \in \Omega_r \colon \mathcal{M}\left(|\nabla u|^q\right) > N_1^q\right\}\right| \geqslant \varepsilon |B_r|,$$

then

$$\Omega_r \subset \left\{x \in \Omega \colon \mathcal{M}\left(|\nabla u|^q\right) > 1\right\} \cup \left\{x \in \Omega \colon \mathcal{M}\left(|\mathbf{f}|^q\right) > \delta^q\right\}.$$

Remark 3.1. The result above is clear from the argument of contradiction. In the sequel, we take N_1, ε and the corresponding δ as given by Lemma 3.3.

Remark 3.2. With the same techniques as in Lemmas 3.1–3.3, we can prove that if u is a weak solution of $-\operatorname{div}(A\nabla u) = \operatorname{div}\mathbf{f}$ in a domain $\Omega \supset 6B$ (B is a ball in $\mathbb{R}^n$), then with similar assumptions as in Lemma 3.3,

$$B \subset \{x \in \Omega\colon \mathcal{M}(|\nabla u|^q) > 1\} \cup \{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \delta^q\}. \tag{3.23}$$

Corollary 3.3. *Assume that u is a weak solution of* (1.1) *in Ω with Ω $(\delta, 48)$-Reifenberg flat, A is uniformly elliptic and $(\delta, 48)$-vanishing. Let $\varepsilon_1 = (\frac{10}{1-\delta})^n\varepsilon$ and k be an integer. Then we have*

$$\big|\{\mathcal{M}(|\nabla u|^q) \geqslant N_1^{(k+1)q}\}\big| \leqslant \varepsilon_1\big(\big|\{\mathcal{M}(|\mathbf{f}|^q) \geqslant (\delta N_1^k)^q\}\big| + \big|\{\mathcal{M}(|\nabla u|^q) \geqslant N_1^{kq}\}\big|\big).$$

Proof. Here we only consider the case $k = 0$, otherwise replace u by $\frac{u}{N_1^k}$ and replace $\mathbf{f}$ by $\frac{\mathbf{f}}{N_1^k}$. Let $G = \{x \in \Omega\colon \mathcal{M}(|\nabla u|^q) \geqslant N_1^q\}$, $H = \{x \in \Omega\colon \mathcal{M}(|\nabla u|^q) \geqslant 1\} \cup \{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) \geqslant \delta^q\}$, then from Lemmas 3.3 and 2.1, we have that

$$\begin{aligned}\big|\{x \in \Omega\colon \mathcal{M}(|\nabla u|^q) \geqslant N_1^q\}\big| \leqslant \left(\frac{10}{1-\delta}\right)^n \varepsilon\big(&\big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) \geqslant \delta^q\}\big| \\ &+ \big|\{x \in \Omega\colon \mathcal{M}(|\nabla u|^q) \geqslant 1\}\big|\big). \qquad \square\end{aligned}$$

Now we begin to prove Theorem 3.1. Simple computation shows that if $\mathbf{f} \in L^\phi$ with $\phi \in \nabla_2$, then $\mathbf{f} \in L^q$ with $1 < q < \min\{n, \gamma\}$, where γ is defined in Definition 2.5. Firstly we claim that

$$I = \sum_{k=-\infty}^{\infty} \big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \delta^q N_1^{kq}\}\big|\phi(N_1^k) \leqslant \frac{C}{\delta^q}\int_\Omega \phi(\mathbf{f})\,dx. \tag{3.24}$$

Since $\phi \in \Delta_2 \cap \nabla_2$, we have

$$\begin{aligned} I &= \sum_{k=-\infty}^{\infty} \big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \delta^q N_1^{kq}\}\big|\phi(N_1^k) \\ &\leqslant C\sum_{k=-\infty}^{\infty} \int_{N_1^{k-1}}^{N_1^k} \big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \delta^q\lambda^q\}\big|\,d\phi(\lambda) \\ &= C\int_0^{\infty} \big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \delta^q\lambda^q\}\big|\,d\phi(\lambda). \end{aligned}$$

For any $\lambda > 0$, define $f_1(x) = |\mathbf{f}(x)|^q$ if $|\mathbf{f}(x)| \geqslant \lambda/2$; $f_1(x) = 0$ otherwise. Then from the weak $(1, 1)$ estimate,

$$\big|\{x \in \Omega\colon \mathcal{M}(|\mathbf{f}|^q) > \lambda^q\}\big| \leqslant \big|\{x \in \Omega\colon \mathcal{M}(f_1) > (1 - 1/2^q)\lambda^q\}\big| \leqslant \frac{C}{\lambda^q}\int_{|\mathbf{f}|>\lambda/2} |\mathbf{f}|^q\,dx.$$

Interchanging the order of integration and integrating by parts, we derive

$$I \leqslant \frac{C}{\delta^q}\int_\Omega |\mathbf{f}|^q \int_0^{2|\mathbf{f}|} \frac{d\phi(\lambda)}{\lambda^q}\,dx. \tag{3.25}$$

Moreover,

$$\int_0^{2|\mathbf{f}|} \frac{d\phi(\lambda)}{\lambda^q} = \frac{\phi(2|\mathbf{f}|)}{(2|\mathbf{f}|)^q} + q\int_0^{2|\mathbf{f}|} \frac{\phi(\lambda)}{\lambda^{q+1}}\,d\lambda. \tag{3.26}$$

The condition $\phi \in \nabla_2$ implies $\frac{\phi(x_1)}{x_1^\gamma} \leqslant \frac{C\phi(x_2)}{x_2^\gamma}$ for all $0 < x_1 < x_2$. Therefore,

$$\int_0^{2|\mathbf{f}|} \frac{\phi(\lambda)}{\lambda^{q+1}}\,d\lambda \leqslant \frac{C\phi(2|\mathbf{f}|)}{(2|\mathbf{f}|)^\gamma}\int_0^{2|\mathbf{f}|} \frac{d\lambda}{\lambda^{q-\gamma+1}} \leqslant \frac{C\phi(f)}{|\mathbf{f}|^q}. \tag{3.27}$$

Combining (3.25) with (3.26) and (3.27), we derive (3.24).

Finally we complete the proof with the following computation:

$$\begin{aligned}\int_\Omega \phi\big(|Du|\big) &\leqslant \int_\Omega \phi\big(\mathcal{M}^{\frac{1}{q}}\big(|Du|^q\big)\big)\,dx \\ &= \sum_{k=-\infty}^{\infty} \int_{N_1^k}^{N_1^{k+1}} \big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|Du|^q\big) > \lambda^q\big\}\big|\,d\phi(\lambda) \\ &\leqslant C\sum_{k=-\infty}^{\infty} \big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|Du|^q\big) > N_1^{kq}\big\}\big|\phi\big(N_1^k\big).\end{aligned}$$

From Corollary 3.3, we have

$$\begin{aligned}&\sum_{k=-\infty}^{\infty} \big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|Du|^q\big) > N_1^{kq}\big\}\big|\phi\big(N_1^k\big) \\ &\quad\leqslant \varepsilon_1 K N_1^\beta \sum_{k=-\infty}^{\infty} \Big(\big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|Du|^q\big) > N_1^{kq}\big\}\big| \\ &\qquad + \big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|\mathbf{f}|^q\big) > \delta^q N_1^{kq}\big\}\big|\Big)\phi\big(N_1^k\big).\end{aligned}$$

Take suitable ε such that $\varepsilon_1 K N_1^\beta = 1/2$, then

$$\sum_{k=-\infty}^{\infty} \big|\big\{x\in\Omega\colon\ \mathcal{M}\big(|Du|^q\big) > N_1^{kq}\big\}\big|\phi\big(N_1^k\big) \leqslant \frac{C}{\delta^q}\int_\Omega \phi\big(|\mathbf{f}|\big)\,dx.$$

That is $\int_\Omega \phi(|Du|)\,dx \leqslant C_1\int_\Omega \phi(|\mathbf{f}|)\,dx$ with C_1 depends on n, ϕ and Ω.

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